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Using Machine Learning to Generate Candidate Globular Clusters in M31

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Abstract

The Andromeda Galaxy (M31) is our closest neighbouring galaxy, and one of the richest sources of information about our own galaxy that we can easily study. The key to this information is its well-studied Globular Cluster (GC) system. GCs are gravitationally bound clusters of stars which date back to the galaxy's accretion period. In this work, we first train an XGBoost classifier on a small GC catalogue, and obtain 72%, 62% and 68% accuracy, precision and recall scores in the best model. We interpret these results using Shapely Additive Explanations (SHAP), and find that positional information is more important than photometry information for this dataset. We then apply this model to unseen astronomical survey data, where it is able to identify promising GC candidates. This work shows that machine learning may significantly aid the GC discovery process, especially with future work into deep imaging surveys through multiple passbands.

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1 Introduction

The Andromeda Galaxy (M31) is our closest neighbouring galaxy, and possibly the most massive in the local group, confirmed as a galaxy in 1929 by Hubble [51, 17]. Research interest in M31 stems from not only the proximity of M31 to our own galaxy, the Milky Way (MW), but also the similarities between M31 and MW. Both are spiral galaxies with similar sizes and ages [16, 53, 46], so studying M31 allows us to draw conclusions about our galaxy, in particular its accretion period and formation.

A specific object of interest to astronomers is the Globular Cluster (GC). GCs consist of around 10^4 to 10^6 gravitationally bound stars, and can often be found in the spherical halo surrounding a galaxy's centre, suggesting that they date back to the early stages of the galaxy's formation [18]. Krauss suggests the median age of GCs to be more than 14 billion years (similar to the age of M31) [22]. Another application of the study of GCs is in information about black holes. Between 10^{-4} and 10^{-6} stars form black holes, so we expect a GC to contain between 0 and 100 black holes. GCs therefore give us a unique insight into the formation and dynamics of black holes [47, 4]. There is also evidence that GCs may be home to the speculated third type of black hole, the intermediate black hole [21].

In M31, the GC system is directly related to the stellar halo, and tend to be more densely populated around regions of high relative density [30, 42, 41]. Recent theories predict that GCs form as a product of intense star-forming episodes, such as those that occur in the highest density region of the galaxy; the disk [13, 23]. The study by Mackey et al. provides evidence for this claim, by showing that the GC distribution correlates strongly with the distribution of low metallicity stars. This implies that GCs can act as tracers for the formation of the galaxy, since low metallicity correlates with older objects [30].

We encounter difficulties in the classification of GCs. One of the most prominent is the misclassification of GCs as either distant galaxies which appear in the background of the target galaxy, or stars which appear in the foreground of the galaxy (i.e. physically located between the point of observation and the target galaxy) [4]. One method to remedy this is to consider the spectroscopy of these objects [36, 5]. This allows us to calculate their radial velocity, using which

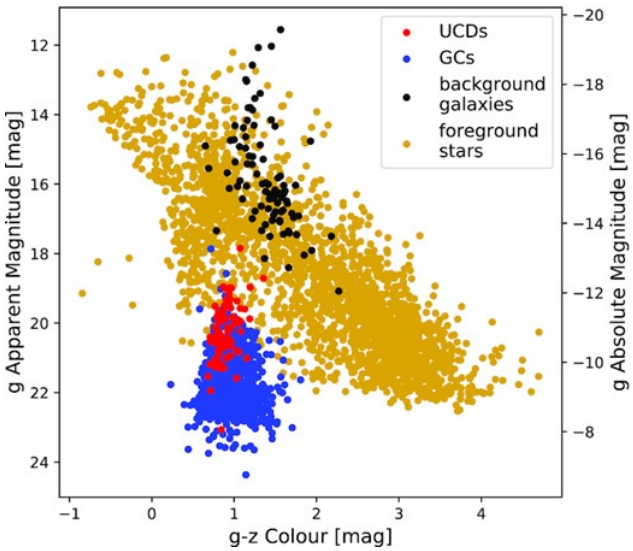

Figure 1.1: Distribution of g-z and g magnitude of various objects in M87 [4]

we can identify GCs associated with a host galaxy. However, this method is slow, meaning that as our technology progresses, and we generate more data, it becomes infeasible. To circumvent this, we look to other methods which may be carried out faster, such as photometry. We define the magnitude as a measure of brightness of an object such that an object of magnitude 1 is 100 times brighter than an object of magnitude 6. Hence a difference of 1 magnitude between objects corresponds to the first object being $\sqrt[5]{100} \approx 2.512$ times brighter than the second. We measure these magnitudes in passbands, which is the range of wavelengths that can pass through a filter. Finally, we define the colour of an object to be the difference in magnitude of the object in two different passbands. For example, the $g-i$ colour of an object is the object's g magnitude, minus its i magnitude [31, 1]. By looking at the photometry of the object, we can see that different types of objects are characterised by their colour. Figure 1.1 (reproduced from Barbisan et al.) shows the clustering in the galaxy M87 of GCs, background galaxies, foreground stars and Ultra Compact Dwarf galaxies (UCDs), which are a class of very compact galaxies with high stellar densities [4, 34]. The classification of GCs is amenable to a machine learning (ML) approach due to the evident clustering of the objects meaning that we may be able to draw a decision boundary between them.

There are a number of examples in the literature that demonstrate the application of ML to GC classification. Brescia et al. in 2012 used a multilayer perceptron to classify GCs in the galaxy NGC 1399, achieving high accuracy and completeness scores of 98.3% and 97.8% respectively on a small dataset using photometry, and show that including structural data, e.g. radius and central surface brightness only slightly improves this result. This shows that the photometry is more important than the structural data in this dataset. In general this suggests that photometry is a useful indicator of the class of an object [9]. Pasquato and Chung used unsupervised clustering techniques to divide GCs into groups [38]. Singlow et al. obtained around 80% accuracy, precision and recall on a 12-class classification problem of objects around M81, using only photometric data, which is impressive considering the number of classes [48]. Recent work

by Askar et al, and Wilson et al have used Naive Bayes classifiers as well as other ML methods such as gradient boosted trees and Support Vector Machines to classify GCs containing black holes, and class-II Young Stellar Objects (a star in the early stages of its formation) respectively [3, 52]. Mohammidi et al. used Localised Generalised Matrix Learning Vector Quantisation and Random Forests to identify UCDs and GCs with a precision and recall of over 93%. These methods allow interpretability by allowing us to calculate feature importance. The authors suggest that $g - r$, $g - i$ and $u - g$ colours are more informative for this type of classification than other colours such as $u - i$ and $i - K$ [35]. Dold and Fahrion report similar scores on a smaller dataset and in the Fornax and Virgo galaxy clusters. A variety of ML methods such as logistic regression, support vector machines, random forests, neural networks and k-nearest neighbours are used and evaluated. The paper presents useful explainable AI figures, such as Local Interpretable Model-agnostic Explanations (LIME) to report that g and z magnitudes are the most important features, once again more important than structural parameters. They find that Random Forests perform the best on tabular data, whereas convolutional neural networks (CNN) perform the best on image data. [12]. There has been other recent research into using CNNs to discover clustering candidates. Wang et al. performed a similar study to the present work, training a CNN on a set of confirmed M31 GCs, and then using the trained model to generated 117 new GC candidates [50]. Finally, Barbisan et al. used ugriz-photometry data to train machine learning models to identify M87 GCs, using random forests and neural networks. This model had many more features than our dataset, making use of (u, g, r, i and z) band magnitudes, as well as $u - g$, $g - r$, $g - i$ and $g - z$ colours, finding flux radii to be of high feature importance [4]. The relevant ML techniques used in these works are explained in section 2.2.2.

Aside from [50], the common aspect these examples lack is the application of the trained models to large scale astronomical data. No model based on photometry has been applied to a large scale catalogue to find GC candidates. In this work, we apply the experimental techniques used in the above literature to train a machine learning model on a small set of confirmed GCs, and optimise its performance, before applying it to a large scale database of photometry data on M31 and the surrounding area. We aim to reduce this large database to a list of plausible GC candidates, which may be then checked visually. This increases the efficiency of the existing process by enlisting the help of ML to reduce the workload when searching for new GCs.

Section 2 of this report will detail the data used in this project, as well as an explanation of the ML techniques we used. In section 3 we analyse different models, choose one to tune and generate our candidate clusters with. We then present an explainable analysis of the model. Section 4 will discuss some of the results in context, explain some data issues we encountered, and finally conclude the project.

2

Methods and Data Set

2.1 Data Set and Preprocessing

2.1.1 General M31 Data

In this project, we use the Pan Andromeda Archaeological Survey (PAndAS) [33]. It surveys more than 400 square degrees of sky around M31 and its close neighbour, the Triangulum galaxy (M33). The data stores the following information about each object: Right Ascension, Declination (both in hexadecimal), i_{CCD} (MegaCam CCD number from 1-36), x_g, y_g (x, y position on CCD in pixels, measured in the g-band),g, d_g (g-band magnitude, and the error in the g-band), i_g (morphology flat in the g-band), x_i, y_i (x, y position on CCD in pixels, measured in the i-band), i, d_i (g-band magnitude, and the error in the i-band), i_i (morphology flat in the i-band), i_a (which is redundant) and Field ID.

In total, PAndAS contains around 97 million observations. We employ various techniques to reduce this number. It is important to remember our goal, which is to identify a candidate list of GCs, which may then be manually checked. We would like our candidate list to contain only GCs, or objects which have a good likelihood of being GCs. Hence we may remove any object from the survey whose morphology (i_g or i_i value) suggests that the object is likely a point source (i.e. a star) (where i_g or $i_i \neq 1$). In addition, we removed any object with significant error in the g or i band magnitude, keeping the rows with both errors less than 0.1 magntiudes. We also removed any object with a g or i magnitude of greater than 23, as this means that the object would be too faint for visual checking. We convert RA and Dec to degrees, a format interpretable by ML. We will only require the RA, Dec, g magnitude and i magnitude from this data set, so remove other features.

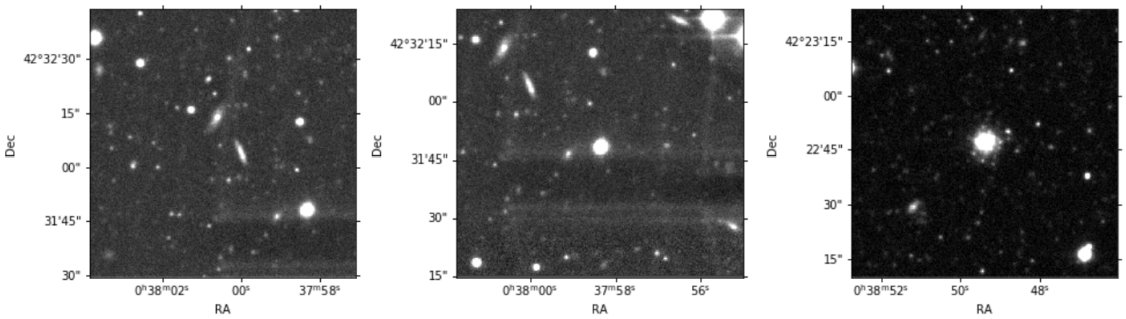

Figure 2.1: Left: 2 galaxies from the RBC, Middle: A star from the RBC, Right: a GC from the RBC

2.1.2 Globular Cluster Specific Data and Training Data

The Revised Bologna Catalogue (RBC) [29] is a catalogue of 2701 objects around M31, their RA, Dec (in degrees), and one of ten flags describing what the object is. Listed are those ten flags, their description, and the number of each in our data in brackets: 0 - no data (0), 1 - GC (727), 2 - Candidate GC (233), 3 - Controversial Object (20), 4 - galaxy (932), 5 - HII region (23), 6 - star(s) (745), 7 - asterism (2), 8 - extended cluster (13), 9 - blank (6). HII-regions are clouds of dust in space, and asterisms are observed patterns of stars in the sky. Figure 2.1 shows an example of a galaxy, which appears more elliptical, a star, which looks like a point source, and a GC, characterised by its bright core and visible outer stars. We see that the majority of the objects are GCs, candidate GCs, foreground stars and background galaxies. We will transform these ten flags into a binary classification problem by labelling GCs and Candidate GCs 1, and all of the other classes 0, except for controversial objects, which we exclude from the data. Binary classification problems are often easier to evaluate than multiclass problems, which is another benefit to this decision. We will see later in the work that intially the model struggles to classify GCs. If we were to frame the problem as multiclass, for example with stars and galaxies as their own categories, we risk the model having an inflated accuracy if it classifies stars and galaxies well, but not GCs. The simplicity of the binary model helps with this.

The RBC is missing the g and i magnitudes required for training the model. Matching the RA and Dec of the RBC to the PAndAS catalog gives a g and i magnitude, which we copy into the RBC. There are some coordinates that appear in the RBC that do not appear in PAndAS. This causes us to have null or zero entries in the extended RBC that we are creating, which will reduce the number of rows in our data set, and potentially affect the performance of the classifier. In the supplied data, we have access only to the g and i magnitudes of objects. This leaves us with only four features in our data set, which may lead to poor performance. To remedy this, we use the 2MASS All Sky Point Source Catalog to access extra magnitude data [49].

2.2 Methods

2.2.1 Evaluation Metrics

In order to assess the effectiveness of ML models, we need evaluation metrics. The accuracy of our model tells us the number of correct classifications over the total number of classifications. However, we must be careful how we interpret this metric, as we find that GCs only account for 27.2% of the training data, meaning that, if the model were to predict that everything is not a GC, then we would obtain an accuracy of 72.8%, a good score for a small dataset. This motivates the use of precision; how often the model is correct when it identifies a GC, and recall, which represents the fraction of GCs that were successfully identified [2]. Another useful representation of these metrics is the confusion matrix, which is a matrix showing the true and false positives and negatives.

The class imbalance described above will require us to be careful with our training, and ensure that our evaluation is equally weighted amongst positive and negative predictions. We will make use of the scikit-learn built in stratified k-fold cross validation [40] in order to achieve this, with $k = 5$. This allows us to monitor the overfitting and underfitting of the model, as well as to ensure that our model will generalise well to the unseen data. The use of cross validation also means that we can include error margains each time that we report a metric score, and average the performance of the model over multiple experiments. In each of the experiments we perform, these error margains will be denoted in brackets next to the result we report (one standard deviation unless stated otherwise). We will plot graphs and other visualisations of the models' performance to aid interpretability of results.

2.2.2 Machine Learning

We consider a range of different Machine Learning models to thoroughly determine which model will be best suited to detecting GCs, described below.

Naive Bayes

The Naive Bayes (NB) classifier makes an often unrealistic assumption that the features are independent given their classification, which greatly simplifies the learning process. Mathematically we can represent this as $P(\mathbf{X}|C) = \prod_{i=1}^{n} P(X_i|C)$, where $\mathbf{X}$ is the feature vector. Suppose we want to learn a function $g(x)$. The NB uses the posterior probabilities given a feature vector as discriminant functions $f_i^*(x) = P(\mathbf{X} = \mathbf{x}|C = i)$ where C is the class of $\mathbf{x}$. Applying Bayes' Theorem we have $P(\mathbf{X} = \mathbf{x}|C = i) = \frac{P(\mathbf{X}=\mathbf{x}|C=i)P(C=i)}{P(\mathbf{X}=\mathbf{x})}$, but $P(\mathbf{X} = \mathbf{x})$ is the same for all classes so we may ignore it, which yields $f_i^*(x) = P(\mathbf{X} = \mathbf{x}|C = i)P(C = i)$. We approximate $g(x)$ with the Bayes classifier $h^*(x) = \underset{i}{argmax} P(\mathbf{X} = \mathbf{x}|C = i)P(C = i)$, which finds the maximum a posteriori probability given $\mathbf{x}$. However, in high dimensional feature space approximating $P(\mathbf{X} = \mathbf{x}|C = i)P(C = i)$ is very difficult, so we use the simplifying assumption of independent feature spaces, so that we the NB Classifier is defined as $f_i^{NB} = \prod_{j-1}^{n} P(X_j = x_j|C = i)P(C = i)$

[43]. This assumption is often violated, especially in our case. We expect, for example, the g and i magnitude to be correlated in our data. Hence this classifier is used as a baseline for the other models, as in [3].

Random Forest

The Random Forest (RF) algorithm combines several randomised decision trees, and aggregates their predictions. In the original paper, Breimann defines an RF as a classifier consisting of a collection of decision trees, $\{h(\mathbf{x}, \Theta_k | k = 1, ..., N\}$, where $\{\Theta_k\}$ are independent and identically distributed random vectors. The classification is made as the most popular class on voting by all of the the classifiers [8]. The N trees are trained on bootstraps of the original dataset, sampled with replacement. A major downside to combining the decision trees is that we lose the interpretability that they give. Despite this, using 'feature bagging' we can obtain individual feature importances for the dataset, adding some interpretability to the model [24]. Figure 2.2 provides a visual representation of the RF training and voting process [20]. Random

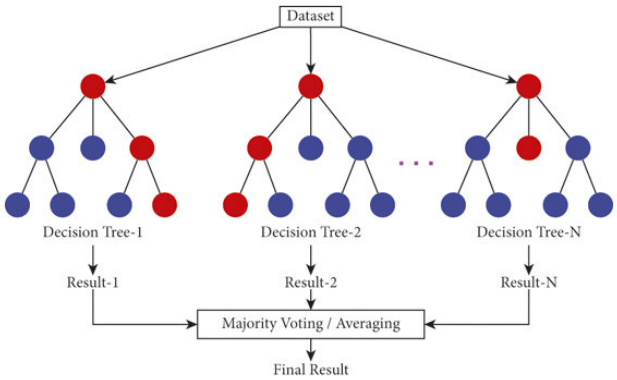

Figure 2.2: Illustration of the random forest algorithm, reproduced from [20]

forests have been shown to be effective in GC detection in the literature, so they are an important choice for our investigation [12, 4, 35].

XGBoost

XGBoost (XGB) is a scalable tree boosting method. Boosting refers to the combination of many weak classifiers into a strong classifier [45]. Gradient boosting requires a differentiable loss function, a weak learner (such as a decision tree), and an additive model which will combine the weak learners [37]. That is, the prediction $\hat{y}_i$ is given by $\hat{y}_i = \sum_{k=1}^{K} f_k(\mathbf{x}_i)$, where f_k are the weak learners. To train the model, we minimise the regularised objective $\mathcal{L}^{(t)} = \sum_i^n l(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)) + \Omega(f_t)$, where l is a differentiable convex loss function of the predicted and true value, and Ω penalises the complexity of the model, which allows for regularisation, which is the benefit of it over traditional gradient boosting models [11]. XGB is the most popular choice of machine learning framework for machine learning competitions [15], but has not been applied in this setting yet, so it is fitting to test it.

Support Vector Machines

Support Vector Machines (SVM) have effectively been used in the literature to classify astronomical objects [3, 12]. The idea behind SVMs is that data that are not linearly separable

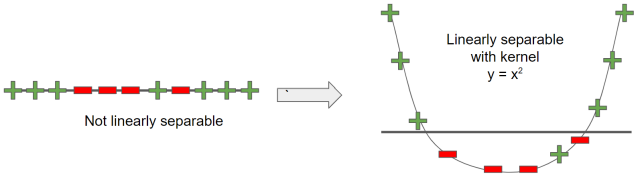

Figure 2.3: The data are not linearly separable in 1D, but the kernel function maps them into a higher dimension where they are more linearly separable.

in low dimensions may be mapped into a higher dimension, where we can find a hyperplane that will separate them. More specifically, if the data is not separable by $\mathbf{y} = \mathbf{w}\mathbf{x} + \gamma$, we may be able to separate it using $\mathbf{y} = \mathbf{w}\phi(\mathbf{x}) + \gamma$, where $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^k$, where d is the original dimension of the data, and $k > d$ is the new feature space [19].

Combined with the previously described evaluation methods, we test each of these models and compare them in the context of recognising GCs. It is important at this stage to define heuristic success criteria for our model. A successful model will:

1. Learn effectively on a small training dataset
2. Miss as few GCs as possible in the testing data
3. Predict large amounts of unseen data quickly

In the next sections, we will refer to these criteria to select machine learning models.

Neural Networks

Some of the above works use neural networks [9, 12, 50, 4]. We decide to exclude neural networks from this study, as they generally work best on large datasets with many parameters [10]. Our dataset is very small, so it is not worth investigating neural networks until we are able to use a much larger dataset.

2.3 Data Exploration

In this section, we present an initial analysis of the training data, in order to guide and motivate our choices of ML architectures. After removing null values, our dataset consists of 1921 rows, 579 of which are GCs (30.1%). Figure 2.4 shows the distribution of g and i magnitudes in the training data. We note some key differences between the distribution of g and i in non GCs against GCs. Table 2.1 shows the important measures of the distribution of the photometry in the g and i bands. We can see that GCs are on average dimmer than non GCs due to their lower mean in the g and i band. The standard deviation in $g - i$ is very similar, as observed in

	Mean			Standard Deviation			Skew		
	g	i	g-i	g	i	g-i	g	i	g-i
GCs	18.97	17.90	1.03	1.77	1.86	0.66	-0.61	-0.63	0.99
Non GCs	19.11	18.05	1.06	1.68	1.56	0.67	-0.69	-0.46	-0.79
Full Data	19.04	17.99	1.05	1.72	1.68	0.67	-0.67	-0.57	-0.12

Table 2.1: Dataset description for the RBC

the histogram in figure 2.4, where we can see a tight distribution in both GCs and non GCs. Also GCs have a lower standard deviation in the g and i band in comparison to non GCs, so extreme values on either extreme are likely to be indicative of a GC. Another key difference is to notice that GCs have a positive skew in $g - i$, whereas the rest of the data has a negative skew. Again this can be seen in the histograms, with the blue peak lying left of the orange. Therefore, objects with a more similar g and i magnitude (and therefore a $g - i$ closer to 0) will be more likely to be classified as a GC. Looking beyond the statistics, we see that there is an obvious gap in the g and

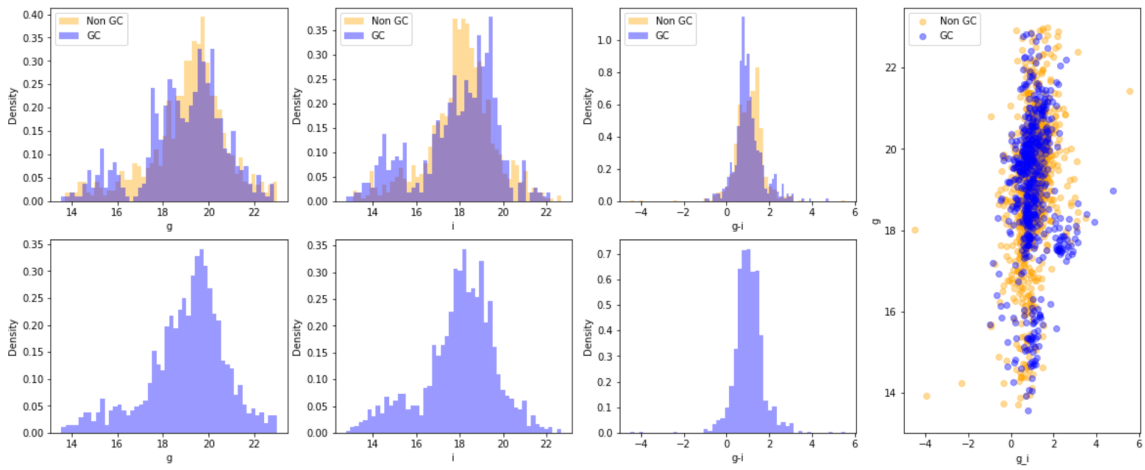

Figure 2.4: Top left row: g, i and $g - i$ distribution in each class of the training data. Bottom left row: Overall g, i and $g - i$ distribution of the training data. Right panel: Colour-magnitude plot for g and $g - i$ of the training data

i histograms around 16 magnitudes. This causes a minor peak around 15 magnitudes in both bands, which can also be observed in the scatter plot of figure 2.4. This peak is not observed in magnitude density plots in the literature, so we are conscious of this anomaly throughout the work. It is likely that this feature will cause some issues with the classification, and potentially produce an inflated number of false negatives around 17 magintudes in the g and i bands, or an inflated number of false positives in the region of the small peak around 15 magnitudes.

3 Results

3.1 Initial Model Analysis

We begin this section with an analysis of ML architectures on the RBC, before we then use the trained model on PAndAS. Table 3.1 shows their mean accuracy, precision and recall, using a 80-20 train-test split and a stratified 5-fold cross validation. In this experiment, each model is used with default parameters. The most promising models are further experimented with and specialised in the coming sections. RF and XGB are the most promising models at this stage

Model	Accuracy	Precision	Recall
Naive Bayes	0.457 (0.06)	0.407 (0.03)	**0.863** (0.09)
Random Forest	**0.712** (0.02)	**0.646** (0.02)	0.542 (0.05)
XGBoost	0.705 (0.01)	0.623 (0.02)	0.576 (0.06)
SVM	0.618 (0.001)	0.000 (0.00)	0.000 (0.00)

Table 3.1: Metrics scores for initial classifiers

performing very similarly across all metrics. Although NB has a very high recall, this comes at the cost of a low accuracy and precision. SVM does not perform well on this dataset, becoming a simple majority class classifier; predicting that everything is not a GC. Different variations on the SVM were then tried (e.g. changing the kernel and the degree of the decision surface), but to no avail. It is interesting that NB has a much higher recall score than every other classifier. We investigate this in the next figure. Recall the aims of the project framed in terms of selecting ML models from section 2.2.2. Criterion 2 suggests that we should favour recall as the most important metric, so perhaps this suggests that we should favour NB as the best classifier at this stage. However, we should also be mindful that the candidates that are selected by the model must be checked by eye. Of course, 100% recall could be obtained by predicting that everything is a GC, meaning our candidate list is just as long as the original list. With this in mind, we begin our deeper investigation of RF, XGB and NB using a confusion matrix. Figure 3.1 shows the

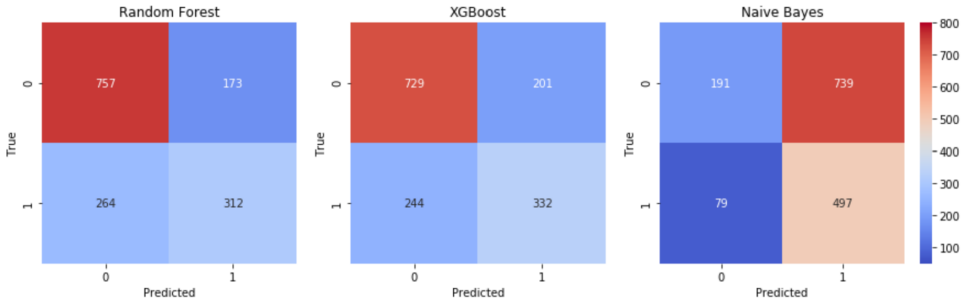

Figure 3.1: Confusion matrices for 5-fold cross validation on RF, XGB and NB

confusion matrix over 5-fold cross validation of our models. The confusion matrices confirm that NB identifies a higher number of GCs correctly than RF and XGB. However, this comes at the cost of a higher false positive rate, reflected in the lower precision score. We also see that XGB identifies more GCs correctly than RF, with a low increase in false positives, so we shall exclude RF from our investigation from this point forth.

3.1.1 Naive Bayes

A quick analysis reveals that Gaussian NB (as we have used above) outperforms the other types of NB offered by sklearn. We attempted to generate a candidate list with Naive Bayes, but it failed to select any candidates. That is, on the training data it became a majority class classifier. Firstly, we expect the performance of NB to be poor, as the features are not independent, for example, g and i have a Pearson correlation coefficient of 0.92. However, this does not explain the lack of candidates. It is known that Naive Bayes classifiers tend to perform worse when the distribution of the training data differs from that of the testing data [14]. In fact, we find that, using a Mann-Whitney U test [32], we find that both the g and i distibutions of the RBC data and the PAndAS data are different at the 1% significance level, so we have evidence as to why NB does not work well when predicting candidates. This is not a surprising result, as the RBC is biased towards GCs, in that there is a higher proportion of GCs in the RBC than in M31 or PAndAS, so the distributions should be different. Because of the underwhelming performance of Naive Bayes in this instance, we disregard it going into the remainder of this work.

3.2 Generating Candidate Clusters with XGBoost

In this section, we generate a first batch of candidate GCs with the standard XGBoost classifier, initially concentrating on the first one hundred fields of the PAndAS catalogue. This concatenated file consists of 4.25 million entries, and the XGB classifier identifies around 114,000 as GC candidates, around 2.7% of the total. A quick analysis shows that the candidates are somewhat promising, with many of the candidates being distant background galaxies, agreeing with the literature as a common object confused with GCs. With these promising results, we fine tune the XGB model in the next section.

3.3 Fine-Tuning XGBoost

This section aims to improve the XGB model we have chosen from the previous stage. We test two ways of doing this. First, we explore the effects of including more features in the data. Then, we tune the parameters.

3.3.1 Inclusion of J, H and K magnitudes

Many similar works use more than two passband magnitudes for their training [4, 9, 35], so in this section we test the effects of including J, H and K magnitudes.. We trained the model fifty times on random 80% training data, 20% testing data splits of the dataset and recorded the accuracy, precision and recall of each run. We then used the Wilcoxon signed rank test at the 5% level of significance, and found that the model excluding J, H and K magnitudes had a significantly higher median in all three metrics than the model including J, H and K magnitudes. We will exclude J, H and K magnitudes from the training data.

3.3.2 Grid Search

A common way to improve the performance of a model is by tuning its hyperparameters. Hyperopt [6] is a python library which allows for parameter optimisation using custom objective functions. In this way, we can find the set of optimal parameters to maximise recall, whilst maintaining a high precision. There are several ways to measure this, one of which is to use the Area Under Receiver Operating Curve (AUC-score). The Receiver Operating Curve (ROC) is a graph of model performance, measured in true positive rate and false positive rate at all possible classification thresholds. It has met praise [25] and criticism [26] in the literature, due to the fact that is a more informative measure than accuracy, but it has drawbacks such as weighting all classification thresholds equally, and ignoring predicted probability values. It also lacks a way to tune the metric to our specific case. Another way is to measure the F_1-score. This is defined as the harmonic mean of precision and recall. We can generalise the F_1 score to the F_β score, where we specify that recall is β times as important to us as precision.

$$F_\beta = (1 + \beta^2) \times \frac{precision \times recall}{\beta^2 \times precision + recall}$$

One of the downsides of F_β score is that it dramatically varies depending on the imbalances in the dataset, as shown by Brabec et al in 2020 [7]. To ensure that we use an acceptable value of β, we use three different values, followed by the AUC score to optimise the XGB parameters, then perform 5-fold cross validation on the training dataset to obtain accuracy, precision and recall scores corresponding to each of these values of β and the AUC metric. The results are shown in table 3.2. The online document for XGBoost (`https://xgboost.readthedocs.io/en/latest/parameter.html#general-parameters`) gives guidance on the parameters, and from this information, we elect to optimise the following parameters in the specified ranges: maximum depth of the weak learners (an integer $\in [1, 15]$), the minimum loss reduction to create

a split in a tree, (a real number $\gamma \in [0, 15]$), the regularisation parameters (an integer $\alpha \in [0, 150]$ and a real number $\lambda \in [0, 5]$), colsample_bytree (a real number $\in [0.5, 1]$ which controls the fraction of columns sampled when building each tree), min_child_weight (minimum sum of instance weight needed in a child, an integer $\in [0, 10]$), and scale_pos_weight (controls the balance of positive and negative classes in the data, a real number $\in [0.2, 2]$. We can clearly

Model number	Optimisation Metric	Accuracy	Precision	Recall
1	$\beta = 0.75$	0.712 (0.03)	0.621 (0.04)	0.627 (0.05)
2	$\beta = 1$	0.717 (0.03)	0.617 (0.04)	0.677 (0.06)
3	$\beta = 1.5$	0.660 (0.03)	0.535 (0.02)	**0.816** (0.04)
4	AUC Score	**0.723** (0.03)	**0.638** (0.04)	0.630 (0.04)

Table 3.2: 5-fold cross validation metric scores after optimisation using the F_β measure

see the effect of altering β in the optimisation; at $\beta = 1.5$ the recall is much higher than the precision, as it has been prioritised in the definition of the optimisation metric. We can make some simple observations, but will need to perform some deeper analysis to pick a final model. Model 4 performs better than model 1, as model 4 performs better in each metric. Model 4 and 2 perform very similarly, but model 2 has better recall. Model 3 does not perform as well in accuracy or precision as any of the other models, but its recall is significantly higher than all of them. Between models 2, 3 and 4, it is difficult to differentiate between them, so we will look further into their performance. The simplest way to do this is to generate a candidate list with each and compare the candidates they produce.

3.3.3 Generating Candidates with Tuned Models

We sampled 10 fields from PAndAS, whose centres are shown in figure 3.2. We chose fields that are from different parts of M31; one from the centre, a few close to the centre, some from the outer halo and one from M33. We then trained each of the models on the whole RBC, and used them to predict the probability that each object in the compiled 10 fields is a GC. We sorted the candidate catalogs first by probability such that the most likely candidates are checked first. Model 4 did not produce good candidates, with most being distant GCs and galaxies. The candidates produced by models 2 and 3 were of similar quality, but model 3 produced nearly 1.5 times as many candidates as model 2, with most extra candidates being stars and galaxies. Model 2 produced some very promising candidates, as shown in figure 3.3.

3.4 Explainable Results using SHAP

Explainable ML is an important area in which there is active research [44]. It allows us to interpret and improve our models better by reducing the 'black box' nature of machine learning. SHapely Additive exPlanation (SHAP) values use a game-theoretic approach to explain the output of machine learning models [27, 28]. For a matrix of objects for the model to predict, SHAP values are calculated in the form of another matrix which explain the influence

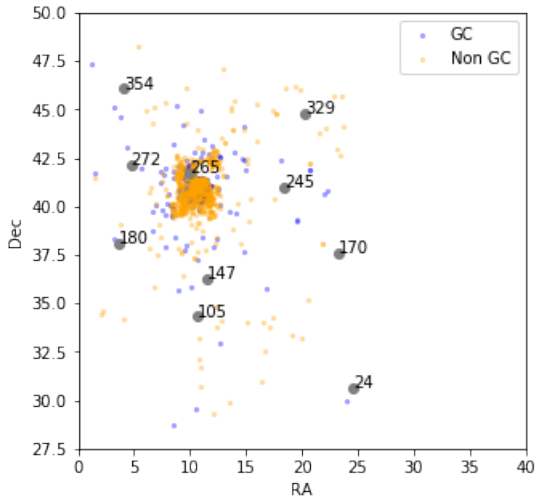

Figure 3.2: The centres of the fields that candidates were generated from. Also plotted is the RBC, with GCs in blue, and Non GCs in orange

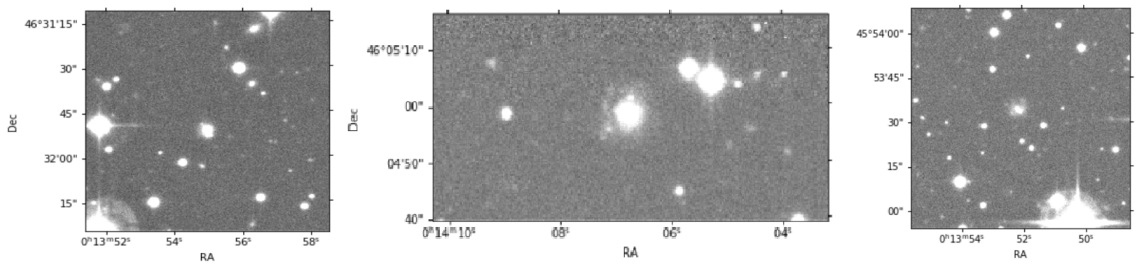

Figure 3.3: 3 GC candidates predicted by model 2

of each feature on the prediction of each entry in the dataset. This means that large swarms of a colour are indicative of many feature values in a similar range having a similar influence on the model. This allows us to view large scale feature influences on the model, as well as small scale information, such as the contributions from each feature on the classification of a specific object. This section will present figures and insights of the SHAP analysis of our model. The first is SHAP's 'beeswarm' plot are shown in figure 3.4. The plot shows each feature, listed in order of total influence on the model (RADEG is the most important, and g is the least important), along with each point in the dataset, coloured by their relative feature value, plotted with their x-coordinate depending on their influence on the model. The further a point is in the positive x direction, the more it influences the model to predict a 1, and the further it is in the negative direction, the more it influences the model to predict a 0. For example, this figure shows that a low RADEG value (indicated by a blue colour) has a large influence on the model being a GC, as it has a high SHAP value. Other observations of this plot help to show why the model struggles to perform on this dataset. In $g - i$, a high $g - i$ is both indicative of a GC, or not a GC, which is contradictory. A similar case is seen for g, where a low g value is indicative of being a GC and not a GC with higher probability. Whilst the tails of the swarm distribution for g magnitudes

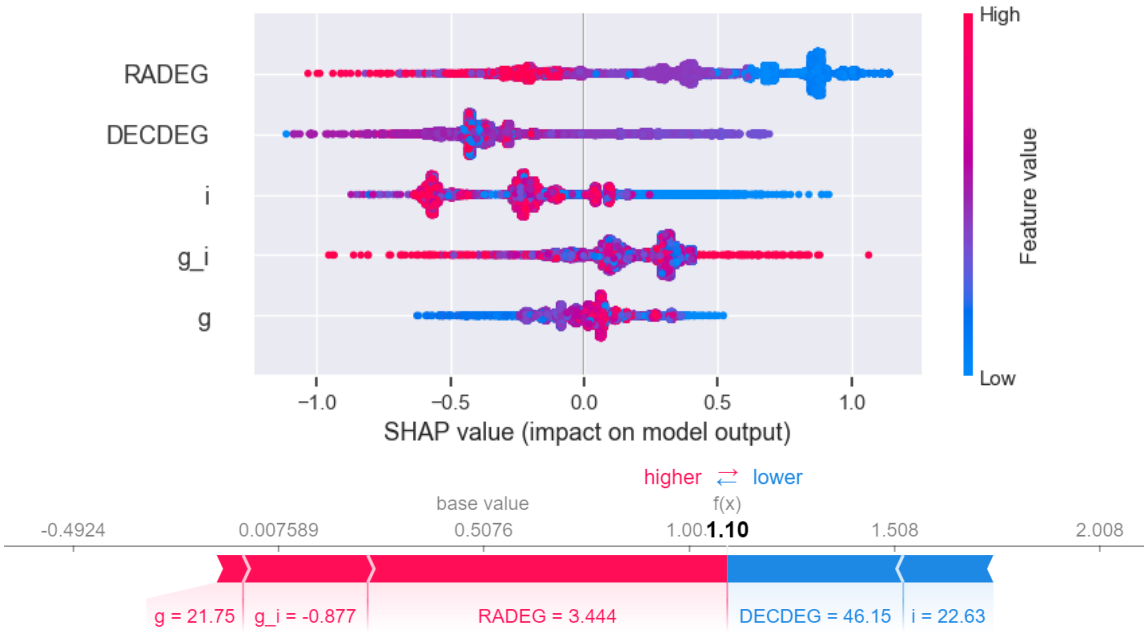

Figure 3.4: Top: Beeswarm plot showing feature importances and the influence that values of each feature have on the model. Bottom: SHAP Force plot to show the individual feature influences of a classification

is purely blue, they do not extend far out to large magnitudes of SHAP value, showing why this feature is not useful for the model. In the i band, we see two large swarms of red that are indicative of a low probability of being a GC, and one smaller swarm which is slightly indicative of the object being a GC. This is in relation to the major peak in the histogram of the i magnitudes (2.4) lying slightly towards the fainter end of the spectrum for GCs as opposed to non GCs. We can also see that low i values tend to be very indicative of a GC. Once again we can draw reference to the histogram of figure 2.4, where the peak around 15 is the cause. If the i value is low, there is a high likelihood that the object belongs to the GC population that dominates this region of the magnitude distribution.

The bottom panel of figure 3.4 shows the influences of each feature for an individual prediction. Specifically, this prediction is for the first GC candidate from figure 3.3. We can see that, as the beeswarm plot says, RADEG gives the most weight to the prediction, suggesting heavily that it should be selected as a candidate. This is supported by the objects $g - i$ and g values, both of which indicate that it should be selected. Interestingly, the high i magnitude is suggesting that the object is not a GC. Perhaps the object belongs to one of the swarms of red in the beeswarm plot. Most likely it belongs to the furthest right large swarm, as its influence is quite small.

4 Discussion

4.1 Classification of Non GCs

By grouping the objects into binary classes, we lose some information when the model predicts a non GC. Figure 4.1 shows bar charts of the true classes of false positives and true negatives (i.e. each time the model predicts a non GC). It shows that the model more often misclassifies objects that are stars rather than galaxies. This is explained by the histograms also in figure 4.1. We can see that the g magnitude distribution of galaxies has a higher mean than the stars, which we expect since many of the galaxies are further away from the point of observation, and so will appear more faint. Often the stars are situated in the MW but appear in the field of view of M31, so will appear brighter. Visually, the distribution of the stars appears to resemble that of GCs more than galaxies. Worse, both the mean and the median of GCs lies in between the mean and median of stars and galaxies. ($\mu_{star} = 18.46 < \mu_{GC} = 18.68 < \mu_{galaxy} = 19.82$ and $Med_{star} = 18.63 < Med_{GC} = 18.67 < Med_{galaxy} = 19.75$). This makes classification difficult, especially seeing that the mean and median values for stars and GCs are so close together. Perhaps it would have been better to create three classes of classification for the model with this hindsight, however we would lose some ease of interpretability, as well as having to deal with the issues discussed in section 2.

4.2 Data issues

4.2.1 Magnitude Distribution in Training Data

There are a number of issues with the data used in this project. Looking at figure 2.4, we can see a small peak in the g and i distribution of the GCs with a mean of around 15. This is a discrepancy with previous studies on GCs, which do not show a bimodal magnitude distribution [39]. In our data preprocessing section, there was an anomaly in processing field m266. After transforming the coordinates to degrees, we ran the same code as used to process

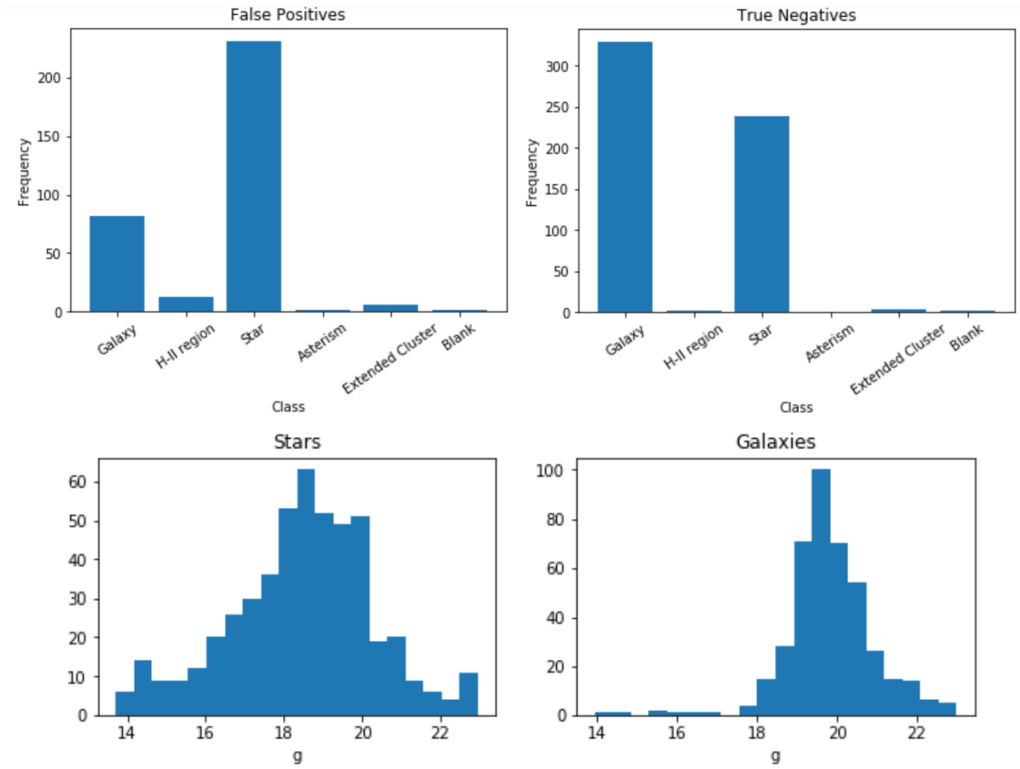

Figure 4.1: Top Left: Bar chart of the RBC class of false positives. Top Right: Bar chart of the RBC class of true negatives. Bottom Left: g magnitude distribution of RBC stars. Bottom right: g magnitude distribution of RBC galaxies

every other field, yet the objects in the resulting file all had the same g and i magnitude. It is not clear whether this is an issue in the PAndAS pipeline, the code used to extract the g and i magnitudes, the astropy match_coordinates_sky function, or a combination of interactions of the three. Regardless, it resulted in the densely populated m266 field being excluded from the dataset, meaning that when the g and i magnitudes were matched to the catalog, we obtained 139 candidates with the same data, and we were forced to exclude these objects from the dataset. It is also not clear whether the objects from m266 would replenish the dip in the histogram, or even if the objects from m266 would improve the quality of our data. This is because nearer the centre of M31, the fields are more dense, as opposed to those in the outer halo. This means that the observations have a high degree of error due to the crowding from adjacent sources. The dip in the histogram does however have an effect on the predictions that the model makes. The histogram for false positives also has this this peak around 15, of which the majority are stars. If we partition the RBC by magnitude (brighter partition $g < 18$, fainter partition $g > 18$), then the model performs better in all three of accuracy precision and recall on the brighter partition. The difference in recall is statistically significant at the 5% level. Initially, this was assumed to be because of the decrease in photometry quality for fainter samples, as shown in [33] figure 8, as sharper imaging quality should lead to better predictions. Whilst this may be true, it could also be explained by the unexpected peak characterising the brighter end of the magnitude distribution.

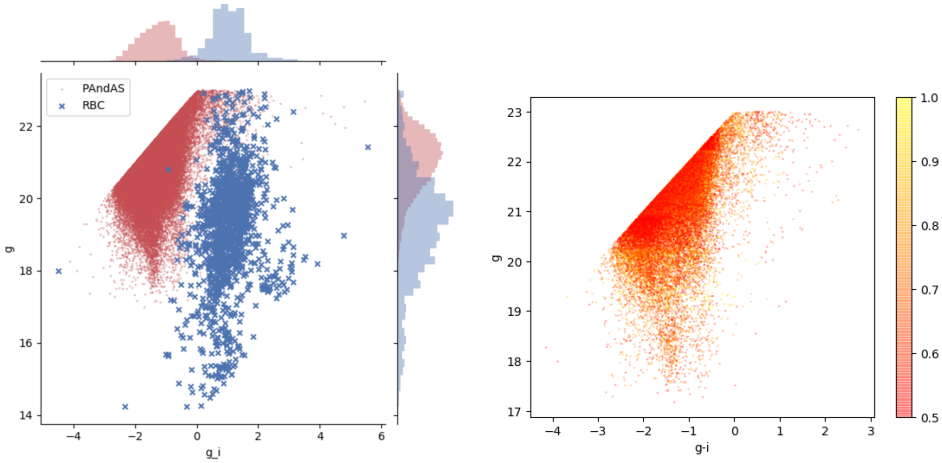

Figure 4.2: Left: The $g - i$ against g colour-magnitude diagram for both the PAndAS fields we generated candidates for and the RBC. Right: The plot of candidates generated by the model, coloured by their probability of being a GC

4.2.2 Discrepancies Between RBC and PAndAS

Another issue with the datasets used is the difference between PAndAS and the RBC. We have already shown that the datasets are statistically different in terms of g and i magnitude, but the differences extend further than that. Figure 4.2 shows a joint scatter plot for the $g - i$ magnitude against the g magnitude of the RBC and the 10 selected PAndAS fields. The sharp boundaries in the diagram are due to the cut-off we set at $g \leq 23$ and $i \leq 23$. We can see that the distributions are almost entirely disjoint, highlighting the major differences between the two datasets. This makes it especially difficult for the model to select the clusters, and would account for the large number of objects it selects that turn out to not be candidates. The right hand panel of figure 4.2 shows the g against $g - i$ colour magnitude diagram for the candidates the final model selected, coloured by the model's predicted probability of them being a GC. We can see that the the vast majority of candidates that had a high probability of prediction were located around the edge of the main cluster, and towards the cluster of the RBC. This is, of course, as expected as the model has learnt the distribution of the GCs in the RBC, and will therefore assign a high probability to those closer to the values that it associated with being a GC. We can see that the distribution of g magnitudes for the PAndAS fields has a higher mean than the distribution for the RBC. This is because the RBC is not a random sample from PAndAS. Especially in earlier studies, brighter objects would have been easier to verify, or even detect in the first place, so the RBC is a sample that is biased towards the brighter magnitude objects.

4.3 $20.2 < g < 21$ Gap in Predictions

We plotted the distribution of the g magnitudes of candidates across different probability thresholds, and found that, at the threshold $p > 0.85$, we observe an unanticipated gap in

the histogram with $20.2 < g < 21$. This gap is not present in the i magnitude distribution until $p > 0.9$. Figure 4.3 shows the difference between the thresholds $p > 0.5$ and $p > 0.85$. There also appears to be a distinct lack of candidates with $g < 19$. The result for $g < 19$ can be explained by the SHAP beeswarm in figure 3.4, which shows that an object with a low g magnitude is either very likely or very unlikely to be a GC, although more often it is unlikely. For an object of mid-range g magnitude however, there is no such explanation. We suspect that the difference is driven by the i magnitude, as this is a more important feature in the prediction than g magnitude, and therefore the gap in the g magnitude distribution is present due to the correlation between g and i magnitude.

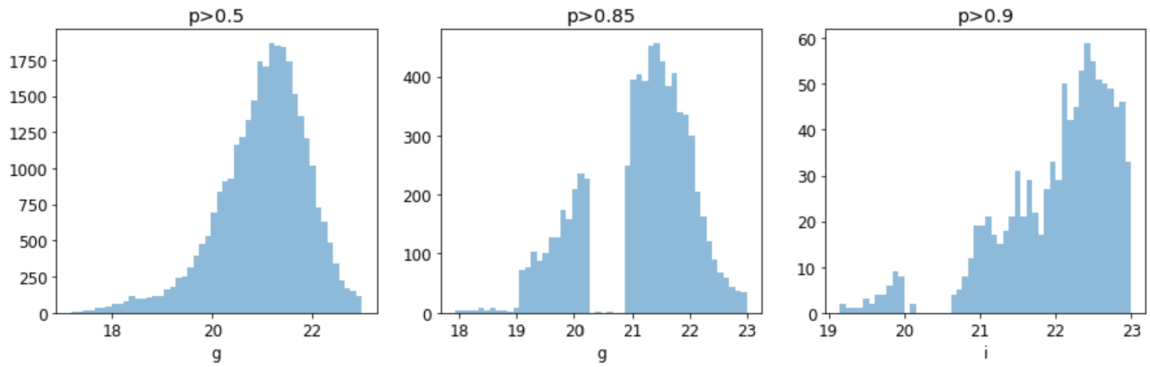

Figure 4.3: Left: Distribution of all candidates selected by the model. Right: Distribution of candidates selected by the model with probability > 0.85

4.4 Limitations and Future Research Directions

The major disadvantage to this model compared to the other works is that our final model performs worse in terms of evaluation metrics. The main reason for this is the size of the dataset. 1921 rows is not large enough to incorporate state-of-the-art neural network models, and it is small enough that non-neural network methods such as the ones we have used in the project will also struggle. In other studies, we see that larger datasets are used. Mohammidi et al use a dataset of around 7500 sources [35], and Barbisan et al use a dataset of around 4000 objects, with a high number of GCs and foreground stars which is useful because, as we have shown, GCs are most often confused with foreground stars [4]. Another limitation of the data is the imbalance in the dataset. The effects of this can be seen in the SVM test, where it became a majority class classifier. A dataset with a more comparable number of GCs against non-GCs would be more favourable for this approach.

As previously shown, the model performs better, and correctly selects more GCs in the brighter end of the magnitude range. We suggested that one reason for this is due to the quality of imaging being higher in that range than in the fainter end. Useful future work may include developing better imaging techniques for fainter objects. Another limitation of this work is that we have only used g and i band magnitudes. Our results could be improved significantly if we had access to large scale astronomical surveys containing colour data for the whole of the

ugriz photmetric system. Whilst we investigated the effect of J, H and K magnitudes, we were unable to obtain u, r and z data for the PAndAS catalogue, which, especially the z band, have been shown to be useful in identifying GCs with ML in other galaxies [4, 35].

Another interesting future research direction is to investigate the possibility of a model which can be used to identify GC candidates in multiple galaxies. It has been shown in the literature that using ML is amenable to the GC systems in many galaxies. Absolute magnitude (M) defines a magnitude system that is independent of distance, and is related to apparent magnitude (m) (the magnitude system we have used) by the equation $M = m - \mu$, where μ is the distance modulus of the object, defined as $5\log_{10}(d) - 5$, where d is the distance to the object in parsecs. If we could train a model on absolute magnitudes rather than apparent magnitudes, we may be able to use this one model to select GC candidates in many different galaxies. This would point to large scale similarities between their GC systems. Of course, a model such as ours would not work as it is heavily dependent on RA and Dec which will dramatically vary from galaxy to galaxy. Perhaps this could be changed for a polar coordinates-like system with the centre of the galaxy treated as the origin, which, mathematically, can encode the same information as the cartesian-like RA/Dec coordinate system. It may also benefit the model greatly if we could include some form of radial distance, as this would help to differentiate between foreground stars and background galaxies.

4.5 Conclusion

In this project, we first evaluated the effectiveness of different ML techniques on small tabular data in classifying astronomical objects in M31. We found that gradient-boosted trees, in particular XGBoost performs the best when this problem is framed as binary classification, achieving reasonable and balanced scores across accuracy, precision and recall metrics. We then used this model on unseen data to select candidate GCs for visual verification, and successfully found four good candidates. Whilst this is not as many candidates as we could have hoped for, it still remains a success as a proof of concept. We supplied ideas for how to improve the model, mostly involving more data in different passbands to try and separate the data. The research into this is motivated by these results, showing that GCs can be identified using a small number of features and a small training catalogue.

References

[1] "8: Astronomical "Color" - Physics LibreTexts". In: (). Accessed: 2022-11-15. URL: %5Curl%7Bhttps://phys.libretexts.org/Bookshelves/Astronomy%5C_%5C_Cosmology/Supplemental%5C_Modules%5C_(Astronomy%5C_and%5C_Cosmology)/Cosmology/Astrophysics%5C_(Richmond)/08%5C%3A%5C_Astronomical%5C_%5C%22Color%5C%22%7D.

[2] Sergio A Alvarez. "An exact analytical relation among recall, precision, and classification accuracy in information retrieval". In: *Boston College, Boston, Technical Report BCCS-02-01* (2002), pp. 1–22.

[3] Ammar Askar et al. "Finding black holes with black boxes–Using machine learning to identify globular clusters with black hole subsystems". In: *Monthly Notices of the Royal Astronomical Society* 485.4 (2019), pp. 5345–5362.

[4] Emilia Barbisan, Jeff Huang, Kristen C. Dage, et al. "Using machine learning to identify extragalactic globular cluster candidates from ground-based photometric surveys of M87". In: *Monthly Notices of the Royal Astronomical Society* 514 (1 June 2022), pp. 943–956. ISSN: 0035-8711. DOI: 10.1093/MNRAS/STAC1396. URL: https://academic.oup.com/mnras/article/514/1/943/6590829.

[5] M. A. Beasley and R. M. Sharples. "Spectroscopy of globular cluster candidates in the Sculptor group galaxies NGC 253 and 55". In: *Monthly Notices of the Royal Astronomical Society* 311.4 (Feb. 2000), pp. 673–682. ISSN: 0035-8711. DOI: 10.1046/j.1365-8711.2000.03080.x. eprint: https://academic.oup.com/mnras/article-pdf/311/4/673/4861780/311-4-673.pdf. URL: https://doi.org/10.1046/j.1365-8711.2000.03080.x.

[6] James Bergstra, Daniel Yamins, and David Cox. "Making a science of model search: Hyperparameter optimization in hundreds of dimensions for vision architectures". In: *International conference on machine learning*. PMLR. 2013, pp. 115–123.

[7] Jan Brabec et al. "On model evaluation under non-constant class imbalance". In: *Computational Science–ICCS 2020: 20th International Conference, Amsterdam, The Netherlands, June 3–5, 2020, Proceedings, Part IV 20*. Springer. 2020, pp. 74–87.

[8] Leo Breiman. "Random forests". In: *Machine learning* 45.1 (2001), pp. 5–32.

[9] Massimo Brescia et al. "The detection of globular clusters in galaxies as a data mining problem". In: *Monthly Notices of the Royal Astronomical Society* 421.2 (2012), pp. 1155–1165.

[10] Lorenzo Brigato and Luca Iocchi. "A close look at deep learning with small data". In: *2020 25th International Conference on Pattern Recognition (ICPR)*. IEEE. 2021, pp. 2490–2497.

[11] Tianqi Chen and Carlos Guestrin. "Xgboost: A scalable tree boosting system". In: *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*. 2016, pp. 785–794.

[12] Dominik Dold and Katja Fahrion. "Evaluating the feasibility of interpretable machine learning for globular cluster detection". In: *Astronomy & Astrophysics* 663 (2022), A81.

[13] Duncan A Forbes, Nate Bastian, Mark Gieles, et al. "Globular cluster formation and evolution in the context of cosmological galaxy assembly: open questions". In: *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* 474.2210 (2018), p. 20170616.

[14] George Forman and Ira Cohen. "Learning from little: Comparison of classifiers given little training". In: *Knowledge Discovery in Databases: PKDD 2004: 8th European Conference on Principles and Practice of Knowledge Discovery in Databases, Pisa, Italy, September 20-24, 2004. Proceedings 8*. Springer. 2004, pp. 161–172.

[15] Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. "Why do tree-based models still outperform deep learning on tabular data?" In: *arXiv preprint arXiv:2207.08815* (2022).

[16] Paul. Hodge. "The Andromeda Galaxy". In: (1992), p. 358. URL: `https://books.google.com/books/about/The%5C_Andromeda%5C_Galaxy.html?id=PUDsCAAAQBAJ`.

[17] Edwin P Hubble. "A spiral nebula as a stellar system, Messier 31." In: *The Astrophysical Journal* 69 (1929).

[18] A P Huxor, N R Tanvir, A M N Ferguson, et al. "Globular clusters in the outer halo of M31: the survey". In: *Mon. Not. R. Astron. Soc* 385 (1989). DOI: `10.1111/j.1365-2966.2008.12882.x`. URL: `https://academic.oup.com/mnras/article/385/4/1989/1033932`.

[19] S. Jothilakshmi and V.N. Gudivada. "Chapter 10 - Large Scale Data Enabled Evolution of Spoken Language Research and Applications". In: *Cognitive Computing: Theory and Applications*. Ed. by Venkat N. Gudivada et al. Vol. 35. Handbook of Statistics. Elsevier, 2016, pp. 301–340. DOI: `https://doi.org/10.1016/bs.host.2016.07.005`. URL: `https://www.sciencedirect.com/science/article/pii/S0169716116300463`.

[20] Muhammad Yaseen Khan et al. "Automated prediction of Good Dictionary EXamples (GDEX): a comprehensive experiment with distant supervision, machine learning, and word embedding-based deep learning techniques". In: *Complexity* 2021 (2021), pp. 1–18.

[21] Bülent Kızıltan, Holger Baumgardt, and Abraham Loeb. "An intermediate-mass black hole in the centre of the globular cluster 47 Tucanae". In: *Nature* 542.7640 (2017), pp. 203–205.

[22] Lawrence M. Krauss. "The age of globular clusters". In: *Physics Reports* 333-334 (4-6 Aug. 2000), pp. 33–45. ISSN: 0370-1573. DOI: `10.1016/S0370-1573(00)00014-4`.

[23] Andrey V Kravtsov and Oleg Y Gnedin. "Formation of globular clusters in hierarchical cosmology". In: *The Astrophysical Journal* 623.2 (2005), p. 650.

[24] Aleksandar Lazarevic and Vipin Kumar. "Feature bagging for outlier detection". In: *Proceedings of the eleventh ACM SIGKDD international conference on Knowledge discovery in data mining*. 2005, pp. 157–166.

[25] Charles X Ling, Jin Huang, Harry Zhang, et al. "AUC: a statistically consistent and more discriminating measure than accuracy". In: *Ijcai*. Vol. 3. 2003, pp. 519–524.

[26] Jorge M Lobo, Alberto Jiménez-Valverde, and Raimundo Real. "AUC: a misleading measure of the performance of predictive distribution models". In: *Global ecology and Biogeography* 17.2 (2008), pp. 145–151.

[27] Scott M Lundberg and Su-In Lee. "A Unified Approach to Interpreting Model Predictions". In: *Advances in Neural Information Processing Systems 30*. Ed. by I. Guyon et al. Curran Associates, Inc., 2017, pp. 4765–4774. URL: `http://papers.nips.cc/paper/7062-a-unified-approach-to-interpreting-model-predictions.pdf`.

[28] Scott M Lundberg et al. "From local explanations to global understanding with explainable AI for trees". In: *Nature machine intelligence* 2.1 (2020), pp. 56–67.

[29] *M31 Globular Clusters*. Accessed: 2022-11-21. URL: `http://www.bo.astro.it/M31/`.

[30] AD Mackey, AMN Ferguson, AP Huxor, et al. "The outer halo globular cluster system of M31–III. Relationship to the stellar halo". In: *Monthly Notices of the Royal Astronomical Society* 484.2 (2019), pp. 1756–1789.

[31] "magnitude | astronomy | Britannica". In: (). Accessed: 2022-11-15. URL: `%5Curl%7Bhttps://www.britannica.com/science/magnitude-astronomy%7D`.

[32] Henry B Mann and Donald R Whitney. "On a test of whether one of two random variables is stochastically larger than the other". In: *The annals of mathematical statistics* (1947), pp. 50–60.

[33] Alan W Mcconnachie, Rodrigo Ibata, Nicolas Martin, et al. "The Large-scale Structure of the Halo of the Andromeda Galaxy. II. Hierarchical Structure in the Pan-Andromeda Archaeological Survey". In: *The Astrophysical Journal* 868 (2018), p. 55. DOI: `10.3847/1538-4357/aae8e7`. URL: `https://doi.org/10.3847/1538-4357/aae8e7`.

[34] Steffen Mieske, Michael Hilker, and Leopoldo Infante. "Fornax compact object survey FCOS: On the nature of Ultra Compact Dwarf galaxies". In: *Astronomy & Astrophysics* 418.2 (2004), pp. 445–458.

[35] Mohammad Mohammadi et al. "Detection of extragalactic Ultra-compact dwarfs and Globular Clusters using Explainable AI techniques". In: *Astronomy and Computing* 39 (2022), p. 100555.

[36] Julie B Nantais and John P Huchra. "Spectroscopy of M81 Globular Clusters". In: *The Astronomical Journal* 139.6 (2010), p. 2620.

[37] Alexey Natekin and Alois Knoll. "Gradient boosting machines, a tutorial". In: *Frontiers in neurorobotics* 7 (2013), p. 21.

[38] Mario Pasquato and Chul Chung. "Clustering clusters: unsupervised machine learning on globular cluster structural parameters". In: *Monthly Notices of the Royal Astronomical Society* 490.3 (2019), pp. 3392–3403.

[39] Mark B Peacock et al. "The M31 globular cluster system: ugriz and K-band photometry and structural parameters". In: *Monthly Notices of the Royal Astronomical Society* 402.2 (2010), pp. 803–818.

[40] Fabian Pedregosa et al. "Scikit-learn: Machine learning in Python". In: *the Journal of machine Learning research* 12 (2011), pp. 2825–2830.

[41] Phillip James Edwin Peebles. "Dark matter and the origin of galaxies and globular star clusters". In: *The Astrophysical Journal* 277 (1984), pp. 470–477.

[42] PJE Peebles and RH Dicke. "Origin of the globular star clusters". In: *The Astrophysical Journal* 154 (1968), p. 891.

[43] Irina Rish et al. "An empirical study of the naive Bayes classifier". In: *IJCAI 2001 workshop on empirical methods in artificial intelligence*. Vol. 3. 22. 2001, pp. 41–46.

[44] Waddah Saeed and Christian Omlin. "Explainable ai (xai): A systematic meta-survey of current challenges and future opportunities". In: *Knowledge-Based Systems* (2023), p. 110273.

[45] Robert E Schapire. "The boosting approach to machine learning: An overview". In: *Nonlinear estimation and classification* (2003), pp. 149–171.

[46] Phil Seiden, Antonella Natta, Renzo Sancisi, et al. "A comparison of the Andromeda and Milky Way galaxies". In: *Symposium - International Astronomical Union* 106 (1985), pp. 423–430. ISSN: 0074-1809. DOI: 10.1017/S0074180900242873. URL: https://www.cambridge.org/core/journals/symposium-international-astronomical-union/article/comparison-of-the-andromeda-and-milky-way-galaxies/0EA5399369BDBA0C784088A41B1B8822.

[47] Steinn Sigurdsson and Lars Hernquist. "Primordial black holes in globular clusters". In: *Nature 1993 364:6436* 364 (6436 1993), pp. 423–425. ISSN: 1476-4687. DOI: 10.1038/364423a0. URL: https://www.nature.com/articles/364423a0.

[48] Tanad Singlow et al. "Identification of Extragalactic Globular Clusters Using Machine Learning Techniques". In: *2022 19th International Conference on Electrical Engineering/Electronics, Computer, Telecommunications and Information Technology (ECTI-CON)*. IEEE. 2022, pp. 1–4.

[49] MF Skrutskie, RM Cutri, R Stiening, et al. "2MASS All-Sky Point Source Catalog". In: *IPAC, doi* 10 (2019).

[50] Shoucheng Wang et al. "Identification of new M 31 star cluster candidates from PAndAS images using convolutional neural networks". In: *Astronomy & Astrophysics* 658 (2022), A51.

[51] Laura L Watkins, N Wyn Evans, and Jin H An. "The masses of the Milky Way and Andromeda galaxies". In: *Monthly Notices of the Royal Astronomical Society* 406.1 (2010), pp. 264–278.

[52] Andrew J Wilson et al. "A naive Bayes classifier for identifying Class II YSOs". In: *Monthly Notices of the Royal Astronomical Society* 521.1 (2023), pp. 354–388.

[53] J. Yin, J. L. Hou, N. Prantzos, et al. "Milky Way versus Andromeda: a tale of two disks". In: *Astronomy Astrophysics* 505 (2 Oct. 2009), pp. 497–508. ISSN: 0004-6361. DOI: 10.1051/0004-6361/200912316. URL: https://www.aanda.org/articles/aa/abs/2009/38/aa12316-09/aa12316-09.html.